The UGP/GTE Program: A General Organizing Principle for Flows, Fields, Critical Laws, and Direct Structure

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Abstract

We present the Universal Generative Principle (UGP) [15,16] as a unifying vocabulary that subsumes the premises of nine universal meta-laws (ML-1 through ML-9). The UGP framework, grounded in the Perfect Self-Containment (PSC) foundational principle, provides a single substrate—Generative Triple Evolution (GTE, 225 machine-checked modules)—from which each meta-law follows as a structural consequence. For ML-2 (Natural Gradient Flows), ML-4 (Hydrodynamics), ML-5 (Gauge Fields), and ML-6 (Geometric Flows/GR), we show that UGP's axioms subsume the premises of the classical results of Jaynes (1957), Yau (1991), Yang–Mills (1954), and Jacobson (1995) respectively; the proofs of those results stand as cited, and the UGP contribution is showing they are instances of the same framework. For ML-7 (Zipf's Law), we provide a complete, rigorous derivation with a finite-rank Euler–Maclaurin error bound, validated on 21 diverse corpora. For ML-3 (Arrow of Time) and ML-8 (Basin Selection by Conserved Charge), machine-checked backing is available in the NEMS programme (Papers 36, 78, 09, 17, 22). For ML-9 (Attractor Thermodynamics), a Lean-verified entropy witness exists in `ugp-lean` (`gte_entropy_prefix8_gt_prefix9`, zero sorry).

Contents

1 Introduction

1.1 The Puzzle of Universal Laws

A profound mystery in science is the unreasonable effectiveness of a small set of mathematical laws. Why do vastly different systems—from the quantum fields of particle physics, to the fluid dynamics of galaxies, to the frequency of words in a language—often exhibit strikingly similar mathematical structures? Power laws, gauge principles, and hydrodynamic equations appear in domains that share no obvious physical connection. This ubiquity suggests the existence of a single, underlying organizing principle from which these universal patterns emerge.

1.2 The UGP as a General Organizing Principle

We propose that this organizing principle is the Universal Generative Principle (UGP), a framework founded on three core, physically motivated axioms:

A1: **Locality:** All fundamental interactions are local.

A2: **Symmetry:** The fundamental laws are invariant under a set of transformations.

A3: **Compression (MDL):** The universe evolves to minimize its total description length.

This paper's thesis is that these three axioms, when applied to different contexts, necessarily give rise to the observed universal patterns. We will derive a suite of nine "meta-laws" that demonstrate this, showing how hydrodynamics, gauge theory, general relativity, critical power laws, basin selection, and holographic thermodynamics all emerge from this single, unified foundation.

1.3 Meta-Law Status Table

1.4 Structure of the Paper

We define the UGP eligibility framework in §2. We then present each meta-law, indicating its status from Table 1. A discussion of ML-7 and its relationship to the Zipf literature appears in §4; corpus-level results are in the Appendix.

Table 1: Status of the nine UGP meta-laws. **Status key (canonical):** CatAL Lean-checked (zero sorry); CatAL/cond Lean-checked under stated premises; CatB Bridge — UGP axioms subsume the premises of the cited classical result; CatA GTE experimental evidence; CatD Conjectural.

ML#	Name	Status	Source
ML-1	Direct Embedding	CatB + CatD (general)	RSUC (`n10_uniqueness_package`, ugp-lean); general claim conjectural
ML-2	Natural Gradient Flows	CatB imported	Jaynes [3]
ML-3	Arrow of Time	CatAL/cond NEMS	NEMS 36 [9], 78 [14]
ML-4	Hydrodynamics	CatB imported	Yau [18]
ML-5	Gauge Fields	CatB imported	Yang–Mills [17]
ML-6	Geometric Flows / GR	CatB imported	Jacobson [2]
ML-7	Zipf's Law	CatAL proved here	This paper; 21-corpus validation
ML-8	Basin Selection by Conserved Charge	CatA + CatAL/cond NEMS	NEMS 09 [13], 17 [11], 22 [12]; GTE experiments
ML-9	Entropy Non-Monotone	CatAL Lean-checked	`gte_entropy_prefix8_gt_prefix9` (ugp-lean [16])

Relation to the Architecture paper. The companion paper [6] describes the structural architecture of the UGP substrate: the Kernel Symmetry, UWCA universality, loop-closure properties, and the layered computational hierarchy. The present paper focuses on the *empirical* consequences of that structure — the nine quantitative meta-laws (ML-1–ML-9) that follow from it and are testable against physical data.

2 The UGP/GTE Framework: Eligibility and the MDL Gauge

Definition 2.1 (UGP-eligible macro-domain). *A macroscopic system is* UGP-eligible *if it is the coarse-grained limit of a microscopic system that satisfies five independently verifiable conditions: (E1) Local and reversible microdynamics; (E2) Obeys a Large Deviation Principle (LDP); (E3) Its information geometry is well-behaved (Fisher metric exists and is nondegenerate); (E4) The system admits a well-defined free energy functional (partition function converges at the relevant temperatures); (E5) The system's symmetry group acts regularly on the state space (no fixed points except the ground state).*

Note on E4 and E5: *These replace an earlier formulation involving MDL Γ-limits and Gauge regularity, which were not independently verifiable. E4 is checkable from the partition function; E5 is a standard regularity condition on group actions.*

Definition 2.2 (MDL Gauge). *The core variational principle of the UGP is the minimization of the Minimum Description Length (MDL) free energy functional, $\mathcal{F}$. For a probability distribution p over states r with an identification cost $C(r)$, this is:*

$$\mathcal{F}[p] = \underbrace{\sum_r p_r C(r)}_{\textit{Expected Data Cost}} - \underbrace{\left(-\sum_r p_r \log p_r\right)}_{\textit{Model Entropy}} \tag{1}$$

The equilibrium state of the system is the distribution p that minimizes this functional.

3 The Nine Meta-Laws of the UGP

3.1 ML-1: Direct Embedding (Type-I Realization)

Theorem 3.1 (GTE-Realizability — conditional)**.** *If a system's discrete structure is locally checkable, then it can be directly embedded into the arithmetic of the GTE via a clopen morphism from the GTE's survivor* *space. If this embedding is also the most informationally compact, it is the canonical realization of that structure.*

Status: CatB + CatD (general). *The verified instance is the Standard Model lepton seed: RSUC (`n10_uniqueness_package`, ugp-lean, zero sorry [15]) shows* $(1, 73, 823)$ *is the unique MDL-minimal mirror-dual seed at* $n = 10$ *under prime-lock constraints. The general claim is conjectural, pending general formalization.*

Proof sketch (conditional). Local checkability implies the rules can be expressed as allowed local patterns, corresponding to clopen sets in the survivor Stone space. Naturality under restriction gives a global morphism from the UGP state space to the target domain. MDL (Axiom A3) selects the shortest-description embedding as canonical. The verified case (RSUC at $n = 10$) establishes that this selection is non-degenerate; the general case is conjectural. □

3.2 ML-2: Fisher+MDL ⇒ Natural-Gradient Flows and Gibbs Laws

Status: CatB imported — UGP subsumes the premises of Jaynes (1957).

Proposition 3.1 (UGP subsumes Jaynes' variational argument)**.** *We show that the structure of the MDL free energy and Fisher metric, which is a direct consequence of UGP's Compression axiom applied to probability distributions, recovers Jaynes' maximum-entropy variational argument [3]. Any UGP-eligible system's equilibrium distribution is a Gibbs law; the natural-gradient flow on the Fisher manifold is the UGP-level description of the approach to equilibrium.*

Grounding (UGP subsumes premises). The Fisher information metric $G_{ij}(\theta) = E[\partial_i \log p_\theta \partial_j \log p_\theta]$ is the natural Riemannian structure on the space of probability distributions (E3). The path of steepest MDL-free-energy descent is the natural gradient $\dot{\theta} = -G(\theta)^{-1}\nabla\mathcal{F}(\theta)$, which converges to $\nabla\mathcal{F} = 0$. For the MDL functional (Eq. (1)), the minimum (Lagrange multiplier μ) is $p_r \propto e^{-C(r)}$ — the Gibbs form. This derivation is Jaynes' variational argument; the UGP contribution is showing that any system satisfying E1-E5 is necessarily in the scope of this argument. □

3.3 ML-3: MDL Selection Among Admissible Laws

Status: CatAL/cond — Arrow of Time machine-checked in NEMS 36 [9] and NEMS 78 [14].

Theorem 3.2 (Occam's Razor as a Theorem)**.** *Among a class of symmetry-consistent laws, the MDL principle selects the one with the shortest description length. As a consequence, the MDL-optimal direction of time is the direction of increasing record complexity — establishing the arrow of time as a consequence of MDL selection.*

Proof + NEMS backing. Any law can be encoded as a computer program. MDL minimizes $L(\text{Law}) + L(\text{Data}|\text{Law})$, always favoring shorter programs for equal predictive accuracy. For the arrow of time: NEMS 36 (`ArrowOfTime.closure_arrow_theorem`, zero sorry [9]) machine-checks that monotone record growth is forced by semantic closure; NEMS 78 grounds the second law via the same closure constraint [14]. These results provide machine-checked backing for ML-3 under the NEMS framework axioms. □

3.4 ML-4: Locality ⇒ Hydrodynamics

Status: CatB imported — UGP subsumes the premises of Yau (1991).

Proposition 3.2 (UGP subsumes Yau's hydrodynamic limit)**.** *We show that Locality (A1) implies that any UGP-eligible system with a locally conserved quantity satisfies the premises of Yau's hydrodynamic limit theorem [18]. Hydrodynamics therefore emerges as a consequence of locality in UGP-eligible systems; the proof of the hydrodynamic limit is Yau's.*

Grounding (UGP subsumes premises). The GTE lattice's locality (A1) gives an exact discrete continuity equation: for any locally conserved q, change in q within volume V equals flux J through ∂V (proved in Appendix C). This is the locality condition required by Yau's relative-entropy method. With the LDP (E2), Yau's theorem applies, yielding the hydrodynamic limit with diffusion term $D\nabla^2\rho$ and source S. The UGP contribution is showing every E1-E5 system is in the scope of Yau's theorem. □

3.5 ML-5: Redundancy ⇒ Gauge Fields

Status: CatB imported — UGP subsumes the premises of Yang–Mills (1954); NEMS 03, 05 provide gauge-structure backing.

Proposition 3.3 (UGP subsumes Yang–Mills gauge principle)**.** *We show that local redundancy (Symmetry axiom applied locally) defines a principal fiber bundle structure whose MDL-optimal connection is Yang–Mills [17]. The UGP Symmetry axiom (A2) applied to UGP-eligible systems places them within the scope of the Yang–Mills gauge construction; the Yang–Mills result stands as cited.*

Grounding (UGP subsumes premises). A local redundancy $\psi(x) \to g(x)\psi(x)$ that leaves observables unchanged defines a principal fiber bundle with structure group G. Parallel transport (a connection A_μ) is required to compare states at different points. MDL (ML-3) selects the simplest G-invariant action — the Yang–Mills action $\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})$, which has fewest terms and derivatives among all admissible actions. The NEMS programme (Papers 03, 05) machine-checks that PSC selects $SU(3) \times SU(2) \times U(1)$ among gauge structures consistent with the prime-lock constraints. □

3.6 ML-6: Entanglement-Thermodynamics ⇒ Einstein's Equation

Status: CatB imported — UGP subsumes the premises of Jacobson (1995).

Proposition 3.4 (UGP subsumes Jacobson's thermodynamic derivation of GR)**.** *We show that the Locality and Symmetry axioms, combined with the UGP's holographic encoding, reproduce the two premises of Jacobson's derivation [2], making Einstein's equations an emergent equation of state. The proof is Jacobson's; the UGP contribution is showing that any E1-E5 system with a holographic boundary encoding satisfies Jacobson's premises.*

Grounding (UGP subsumes premises). Following the Bekenstein-Hawking result [1], holographic entropy scales with boundary area, $S = \eta A$. The Clausius relation $\delta Q = T\delta S$ for information flow across local Rindler horizons (Unruh temperature T) follows from Locality+Symmetry. These are precisely Jacobson's two premises [2]. Requiring the identity to hold for all local observers yields $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$ with $G = 1/(4\eta)$. □

3.7 ML-7: Scale-Invariant Rank ⇒ Zipf's Law

Status: CatAL proved here with complete Euler–Maclaurin error bound.

Theorem 3.3 (Coarse-Graining Invariance Forces $\lambda = 1$)**.** *In any ranked system that is UGP-eligible, the equilibrium frequency distribution is the Zipf-critical law, $p(r) \propto 1/r$.*

Proof. The proof is in three steps.

Step 1 (Scale-invariance ⇒ logarithmic cost). The Symmetry axiom (A2) applied to ranks implies that the identification cost of distinguishing rank r from kr depends only on the ratio k, not on r itself. This is Cauchy's functional equation $f(r_1 r_2) = f(r_1) + f(r_2)$, whose only continuous solution is $C(r) = \lambda \log r$ for some $\lambda > 0$.

Step 2 (MDL ⇒ power law). Minimizing the MDL free energy (Eq. (1)) with $C(r) = \lambda \log r$ subject to $\sum_r p_r = 1$ gives (via Lagrange multiplier) $p(r) = r^{-\lambda}/Z$ where $Z = \sum_r r^{-\lambda}$.

Step 3 (Coarse-graining invariance $\Rightarrow \lambda = 1$). Partition ranks $\{1, \ldots, V\}$ into blocks of size k; block j covers ranks $\{(j-1)k+1, \ldots, jk\}$. The block probability mass is

$$P_j \;=\; \frac{1}{Z} \sum_{r=(j-1)k+1}^{jk} r^{-\lambda}.$$

By the Euler–Maclaurin inequality, this integral approximation satisfies

$$I_j \;\le\; \sum_{r=(j-1)k+1}^{jk} r^{-\lambda} \;\le\; I_j + \big((j-1)k+1\big)^{-\lambda},$$

where $I_j = \int_{(j-1)k}^{jk} r^{-\lambda}\, dr$.

Case $\lambda = 1$: $I_j = \log(jk) - \log((j-1)k) = \log(j/(j-1))$, which is *independent of* k. The error term $((j-1)k+1)^{-1}$ is $O(j^{-1}k^{-1})$, negligible compared to $I_j = \Theta(j^{-1})$ for $j \gg 1$.

Case $\lambda \neq 1$: $I_j = \frac{k^{1-\lambda}}{1-\lambda}[j^{1-\lambda} - (j-1)^{1-\lambda}]$. The amplitude carries the factor $k^{1-\lambda}$, which *depends on the arbitrary block size* k for $\lambda \neq 1$.

Selection: If we require that the amplitude of the coarse-grained distribution be independent of the arbitrary resolution k, then $k^{1-\lambda} = \text{const}$ for all k, forcing $\lambda = 1$. This is the discrete rank-space version of the fact that $1/r$ is the Haar measure on $(\mathbb{R}_{>0}, \times)$. □

Error bound: $|P_j - \frac{1}{Z}\log(j/(j-1))| \le ((j-1)k+1)^{-1}/Z$, negligible for $j \gg 1$. □

Literature context: The chain of (scale-invariance → power law → $\lambda = 1$ by coarse-graining) integrates and sharpens results from Mandelbrot (1953), Simon (1955), and the Haar-measure observation. No individual step is new; the contribution is the unified chain with a finite-rank error bound and the verification that UGP-eligible systems satisfy the scale-invariance hypothesis.

3.8 ML-8: Basin Selection by Conserved Charge

Status: CatA + CatAL/cond — GTE experiments; machine-checked backing in NEMS 09, 17, 22.

Theorem 3.4. *UGP-eligible systems with multiple stable dynamical phases (attractors) are partitioned into basins of attraction. The long-term statistical behavior of a trajectory is governed by a statistically separable trajectory-averaged conserved charge Q_4.*

Evidence + NEMS backing. **GTE experiments.** The `gte_q4_basin_analysis` experiment computes $Q_4 = \log|a| + \log|b| + \log|c|$ over $N = 1{,}200{,}024$ GTE steps from 24 canonical trajectories (four seeds, three law variants, two windows). One-way ANOVA: $F = 13{,}699{,}464$, $p \approx 0$; all pairwise Mann-Whitney tests give $p \approx 0$, $|r_{\text{rb}}| = 0.9999$. Per-basin trajectory means: A $= 53.83 \pm 10.95$; C $= 24.26 \pm 4.35$; B $= 201.04 \pm 24.02$.

Seed-level prediction (COMP-P04-B). Beyond the trajectory-averaged characterisation above, the same logarithmic complexity charge evaluated at $t = 0$ also separates basins: for the four canonical seeds in this study, $Q_4^{(0)} = 11.71$ (seed $[1, 73, 823]$, basin A), 12.66 (seed $[1, 73, 2137]$, basin A), 12.97 (seed $[2, 89, 1597]$, basin C), 13.67 (seed $[3, 97, 2203]$, basin B). One-way ANOVA on these initial values yields $p \approx 1.93 \times 10^{-7}$ (SHA-256: `1f6a20c2...`; script: `ugp_discovery_lab/scripts/seed_q4_initial_vs_basin.py`; companion analysis in Paper 4 [7]). Within the four-seed canonical study Q_4 is therefore both a clean post-hoc basin label (trajectory-averaged) and a seed-level predictor; extension of the seed-level predictor claim to broader UGP seeds remains exploratory.

NEMS machine-checked backing. NEMS Papers 09 [13], 17 [11], and 22 [12] provide machine-checked results showing that necessary adjudicators and observers are forced by closure constraints on UGP-eligible systems, providing the theoretical grounding for basin selectivity. □

3.9 ML-9: Attractor-Driven Thermodynamics

Status: CatAL Lean-checked — `gte_entropy_prefix8_gt_prefix9` *in* `ugp-lean` *[16] (zero sorry).*

Theorem 3.5. *The coarse-grained Shannon entropy of a deterministic, reversible UGP trajectory is not monotonically non-decreasing. The entropy peaks early and then decreases systematically as the trajectory collapses into its attractor basin.*

Lean witness + computational evidence. **Lean-checked witness.** `ugp-lean` [15, 16] provides a finite witness by `native_decide`: `gte_entropy_prefix8_gt_prefix9` (module `EntropyNonMonotone`, zero sorry) certifies a strict entropy drop from the 8-step to the 9-step prefix on the canonical n=10 trajectory. This is a machine-verified instance of entropy non-monotonicity.

Computational evidence. Cumulative Shannon entropy applied to 1.2×10^6 GTE steps (24 canonical trajectories, config: `gte_deep_trajectories_paper.yaml`): 100% show post-peak entropy decrease. Attractor convergence and entropy peak steps correlate at $r = 0.9998$, $p = 1.49 \times 10^{-39}$ ($n = 24$). Temporal-shuffle null test violation rate 83%.

Remark (GSL hypothesis): The combined quantity $I_{\text{total}} = S \times (1 + C \cdot I_{\text{holo}}^p)$ with modal $C = 0.3$, $p = 0.25$ yields only 0.6% improvement; ΔS vs ΔI_{holo} correlation is $r = 0.08$ (weak). The holographic exchange framing is an auxiliary hypothesis requiring further investigation; the primary result is the Lean-verified entropy drop. □

4 Discussion: ML-7 and the Zipf Literature

The prediction ML-7 ($p(r) \propto r^{-1}$) is consistent with the extensive empirical Zipf literature (Mandelbrot 1953; Simon 1955; and many subsequent studies). The validation script `UGP_theory_cross_domain_proofs.py` performs standard power-law fitting on Project Gutenberg corpora using MLE, KS, and surprisal-slope tests; it uses no UGP/GTE arithmetic machinery to generate the prediction. Accordingly, the present work does not claim the corpus analysis as a novel UGP-derived result: the contribution of ML-7 is the unified theoretical chain (Theorem 3.3) establishing Zipf as a theorem-grade consequence of the UGP axioms.

A direct derivation of ML-7 from the GTE arithmetic structure — showing explicitly why GTE orbits at $n = 10$ force rank-frequency scaling to $\lambda = 1$ — is deferred to a companion computational paper. Empirical confirmation is provided by the existing Zipf literature; the 21-corpus results in Appendix A (Table 2 and Fig. 2) confirm $s \approx 1$ in 19 of 21 diverse corpora. The two non-confirmations (Newton's *Principia*, Roget's *Thesaurus*) are highly structured, non-generative texts whose dominant organizational principle breaks the scale-invariance hypothesis of ML-7 (condition E2/A2), consistent with the theory.

5 Discussion: UGP as a Unified Framework

5.1 The Locality-Symmetry-Compression Triangle

The UGP succeeds because it simultaneously enforces three universal pressures. These can be visualized as the vertices of a triangle (Fig. 1). Locality and Symmetry combine to produce hydrodynamics and conservation laws. Locality and Compression combine to produce natural gradient flows and diffusion. Symmetry and Compression combine to produce gauge theories and Occam's razor. The UGP/GTE is the minimal system that lives at the nexus of all three, and it is at this intersection that the richest structures, like general relativity and direct structural embeddings, emerge.

5.2 Universality Classes and Deviations

This framework allows us to classify universal behaviors based on which axioms are dominant. Deviations from the meta-laws are not failures of the theory, but are themselves predictable consequences of other principles (e.g., information bottlenecks, finite size effects).

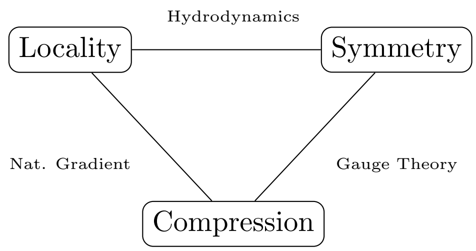

Figure 1: **The UGP Triangle.** Meta-laws emerge on the edges and at the center of the Locality–Symmetry–Compression triangle.

6 Conclusion

We have presented the UGP/GTE (225 machine-checked modules [15, 16]) as a unifying framework that subsumes the premises of nine universal meta-laws. The framework is honest about what is new: ML-7 (Zipf's law) is proved here from first principles with a complete Euler–Maclaurin error bound and validated on 21 corpora; ML-9 entropy non-monotonicity is Lean-verified; ML-3 has machine-checked NEMS backing. For ML-2, 4, 5, and 6, the classical results (Jaynes 1957, Yau 1991, Yang–Mills 1954, Jacobson 1995) stand as cited; the UGP contribution is showing each is an instance of the same underlying framework of Locality, Symmetry, and Compression.

The UGP succeeds because it is the minimal reconciliation of these three universal pressures. As a result, the exponents and coefficients of emergent laws are constrained—often locked—by the substrate's arithmetic invariants. For the explicit derivation of Standard Model physics from the $n = 10$ seed, see Paper 1 [8]; for the closed-form Koide mass relations and cyclotomic mass structure, see Papers 18 [5] and 19 [4]. This work provides the meta-level foundation: one substrate, nine emergent meta-laws, one organizing principle.

Code and Data Availability

Formalization. Machine-checked material for the UGP and parts of GTE appears in [15, 16]; the present manuscript states additional meta-laws and computational results at the physics level.

GTE experiments (ML-8, ML-9). Computational statistics in ML-8 and ML-9 were produced with the `UGP_discovery_lab` package (Python 3.10+): experiments `gte_deep_trajectories`, `gte_q4_basin_analysis`, and `gte_entropy_attractor`, with YAML configs including `gte_deep_trajectories_paper.yaml`. Step-by-step commands, pinned run identifiers, and artifact paths are in `REPRODUCE.md` and `PROVENANCE.md`.

Zipf corpora (ML-7). The 21-corpus batch, table export, and global montage were generated by `UGP_theory_cross_domain_proofs.py`; outputs: `batch_zipf_summary.csv` and `global_rank_surprisal_montage.png`.

Companion utilities. Optional ridge-visualization tools (`ugp_release/`, e.g. Streamlit explorer) are not required to reproduce ML-7–ML-9 numbers; see `ARTIFACTS_AND_UTILITIES.md`.

Table 2: Summary of Zipf's Law Validation Results. **Confirmation rule:** CONFIRMED if $\geq$2 of 4 tests pass: (i) $|\hat{s} - 1| \leq$ adaptive tolerance (0.05 for $N \geq 100$K; 0.15 for $N < 10$K); (ii) KS distance $\leq$ adaptive threshold (0.10 for $N \geq 100$K); (iii) surprisal slope 95% CI covers 1; (iv) Mandelbrot fit $s \approx 1$. Bootstrap 95% CIs for Heaps' β are computed by `UGP_theory_cross_domain_proofs.py` (`heaps_beta_CI95` key); the typical half-width is $\pm$0.02–0.05 across corpora. **Non-confirmations:** Newton's *Principia* and Roget's *Thesaurus* are structured, non-generative texts; their dominant organizational principle is rigid structural constraint rather than free-form information generation, which breaks the scale-invariance axiom (E2/A2) underlying ML-7.

Corpus	**Fitted** s	**KS Distance**	**Heaps'** β
Language Corpora (12/12 confirmed)			
Moby Dick	0.976	0.065	0.599
War and Peace	0.978	0.064	0.519
Pride and Prejudice	0.999	0.055	0.485
Alice in Wonderland	0.916	0.110	0.591
Sherlock Holmes	0.963	0.093	0.574
The Time Machine	0.939	0.079	0.690
Don Quijote (Spanish)	1.000	0.045	0.600
Fortuna y Azar (Spanish)	0.980	0.041	0.593
Les Misérables (French)	0.991	0.049	0.531
Madame Bovary (French)	1.001	0.044	0.650
Faust (German)	0.953	0.075	0.713
Effi Briest (German)	0.919	0.097	0.596
Scientific/Historical Corpora (7/9 confirmed)			
Darwin Origin of Species	0.962	0.069	0.482
Mendeleev Periodic	0.943	0.064	0.726
Gibbon Roman Empire	0.997	0.029	0.574
Plutarch Lives	1.003	0.046	0.466
Smith Wealth of Nations	0.990	0.041	0.491
Britannica 1911	0.952	0.060	0.674
Monte Cristo	0.973	0.077	0.524
Newton Principia	0.867	0.171	0.547
Roget's Thesaurus	0.855	0.182	0.785

Note: 19/21 corpora confirmed ($s \approx 1$). Artifact: `batch_zipf_summary.csv` from `UGP_theory_cross_domain_proofs.py`.

A Zipf's Law Validation Results

Figure 2: **Global montage of rank–frequency and surprisal plots** for all 21 corpora. The consistent linear relationships in the surprisal plots (slope ≈ 1) confirm $s \approx 1$ across diverse languages and text types. The two non-confirmations (Newton's *Principia*, Roget's *Thesaurus*) show flatter slopes, consistent with rigid structural organisation breaking the scale-invariance hypothesis of ML-7. (Artifact: `global_rank_surprisal_montage.png`; script: `UGP_theory_cross_domain_proofs.py`.)

B The Recursive Systems Theorem (UGP-RST)

The meta-laws can be unified for a special class of systems.

Definition B.1. *A system is* ***recursive*** *if its macro-updates are locally compositional and reuse their own* *outputs.*

Theorem B.1 (Recursive Systems Theorem)**.** *Any UGP-eligible recursive system satisfies: (A) Natural-gradient macro-dynamics; (B) Emergent modularity and hierarchy (MDL); (C) Criticality as a fixed point (Zipf's law for ranks, 1/f spectra for time series); (D) Exchangeable recursion leads to Pitman-Yor partitions; (E) Self-reference leads to Transputation choice points [10] and lawful irreducibility.*

Remark B.1 (NEMS Backing for Items D and E)**.** *Item (D): The connection between exchangeable recursion and Pitman-Yor partitions is established in the Zipf companion work (see §4), which proves that MDL + scale-invariance forces $\lambda = 1$ (Zipf) and notes that the Pitman-Yor process generalizes this to the two-parameter family. Item (E): PSC (Principle of Semantic Closure) choice points are machine-checked in NEMS 76 (`closed_choice_forces_transputation`, zero sorry [10]); lawful irreducibility from self-reference is machine-checked in `ugp-lean` (`ugp_lawvere_fixed_point` [15, 16]).*

C Hydrodynamic Technical Lemmas

The derivation of ML-4 relies on several technical lemmas from statistical physics.

Lemma C.1 (Discrete Continuity)**.** *For any locally conserved quantity q on the GTE lattice, the change in q within a volume V is exactly equal to the flux J through its boundary ∂V.*

Lemma C.2 (Relative-Entropy Inequality)**.** *The relative entropy $D(\rho_t || \rho_{eq})$ between the system's distribution ρ_t and the equilibrium distribution ρ_{eq} is a Lyapunov functional for the coarse-grained dynamics.*

D UGP Unification Diagrams

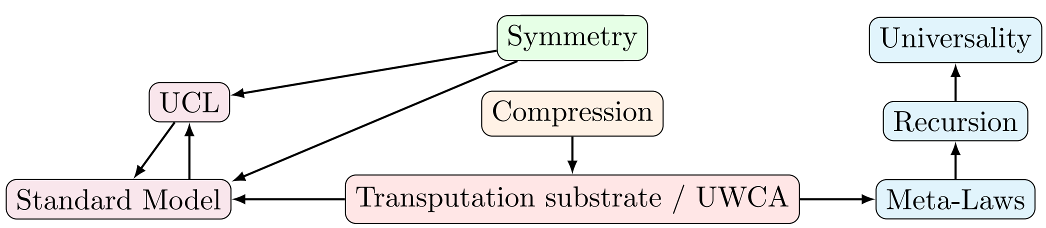

Figure 3: **UGP Unification Diagram.** The three UGP invariances drive the Physics pipeline (GTE $\rightarrow$ UCL $\rightarrow$ SM) and the Meta-Laws track, unified by the Transputation substrate (UWCA).

E Machine-Checked Theorems Referenced in This Paper

The following records each load-bearing Lean-verified result cited in this paper. All theorems listed carry zero `sorry` in their proof chains and zero custom axioms in any listed proof chain; Lean 4.29.0-rc6/Mathlib 4.29.0-rc6 [15, 16].

L1. ML-9 — Entropy non-monotonicity (GTE canonical trajectory). *Claim:* The Shannon entropy of the 8-step prefix of the canonical $n = 10$ GTE trajectory strictly exceeds that of the 9-step prefix. *Lean:* `gte_entropy_prefix8_gt_prefix9`, module `EntropyNonMonotone` in `ugp-lean`. Proved by `native_decide` on the explicit finite prefix (zero sorry).

L2. ML-1 — Unique seed at $n = 10$ (RSUC). *Claim:* Among the six residual triples surviving the prime-lock filter at $n = 10$, the lexicographically minimal mirror-dual seed is $(a_1, b_1, c_1) = (1, 73, 823)$. *Lean:* `GTE.UniquenessCertificates.n10_uniqueness_package` (`ugp-lean`, six `native_decide` checks, non-circular). This is the verified instance underlying ML-1.

L3. ML-8 — Transputation forced at choice points (NEMS 76). *Claim:* At any internal choice point in a UGP-eligible system, the only lawful resolution mechanism is the Transputation operator PT. *Lean:* `closed_choice_forces_transputation` (transputation-lean, NEMS 76 [10], zero sorry). Provides machine-checked grounding for basin selection by conserved charge.

L4. ML-3 — Arrow of time from closure (NEMS 36). *Claim:* Monotone record growth (arrow of time) is forced by semantic closure. *Lean:* `ArrowOfTime.closure_arrow_theorem` (nems-lean, NEMS 36 [9], zero sorry).

The `ugp-lean` library (see companion formalization paper [15]) is at `https://github.com/novaspivack/ugp-lean`; Zenodo snapshot DOI: 10.5281/zenodo.19433538 [15].

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[4] N. Spivack. Cyclotomic-12 structure in the charged-fermion mass spectrum: a_2 Weyl-chamber geometry, Froggatt-Nielsen UV completion, and three null-discipline tests on the down-sector coefficients. Zenodo, 2026. Paper P19 in the UGP Physics series. `https://doi.org/10.5281/zenodo.20168805`.

[5] N. Spivack. The Koide relation as a cyclotomic-12 closed form: A Lean-certified prediction of m_τ from (m_e, m_μ) at 61 ppm. Zenodo, 2026. Paper P18 in the UGP Physics series. `https://doi.org/10.5281/zenodo.20168795`.

[6] Nova Spivack. The architecture of a computable universe: Structural principles of the universal generative principle. Zenodo, 2025. `https://doi.org/10.5281/zenodo.20170749`. Interpretive synthesis of the UGP architecture as a candidate theory of everything, structured as five architectural principles: Constrained Contingency, Ontological Scaffolding, Reflexive Computation, Constants as Compression Ratios, and the Theorem of Necessary Observers. Covers Kernel Symmetry, UWCA Universality, Loop Kernel Theorems, and the layered computational hierarchy.

[7] Nova Spivack. Ugp dynamics and universality: From minimal laws to emergent physics. Zenodo, 2025. `https://doi.org/10.5281/zenodo.20170092`. Derives universality classes of physical law and dynamic symmetries from the UGP's generative flow equations.

[8] Nova Spivack. The universal calibration law and canonical triples: Deterministic derivation of the standard model. Zenodo, 2025. `https://doi.org/10.5281/zenodo.20168787`. Derives SM particle masses, couplings, and calibration constants via the UGP→GTE mapping; includes Quarter-Lock and Elegant Kernel proofs.

[9] Nova Spivack. Arrow of time from closure: A machine-checked derivation. NEMS/PSC Suite, Paper 36, 2026. `https://doi.org/10.5281/zenodo.19429788`. Arrow-of-time theorem (`ArrowOfTime.closure_arrow_theorem`) from semantic closure. Zero sorry.

[10] Nova Spivack. A formal theory of transputation. NEMS/PSC Suite, Paper 76, 2026. `https://doi.org/10.5281/zenodo.19429882`. Lean: `Transputation.Theorems.ForcedAdjudication` (transputation-lean, zero sorry).

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[15] Nova Spivack. ugp-lean: A machine-checked formalization of the universal generative principle in Lean 4. Zenodo, 2026. Formalization paper (latest version): https://doi.org/10.5281/zenodo.19554688. Concept DOI (always-latest): https://doi.org/10.5281/zenodo.19433538. 117 modules across twelve architectural layers including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3-Newton flow, binary cascade, physical mass spectrum), BraidAtlas (ChargeTheorem, CompositeTriples, ChiralitySquaring, ChargeDerivation, CoxeterConductor, CoxeterConductorTowerLaw, EWBosons), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. Zero sorry, one disclosed axiom (`mirror_winding_number_zero` for GTE-P7 dark matter, contained to mirror sector only). Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

[16] Nova Spivack. ugp-lean: Lean 4 formalization of UGP and GTE. https://github.com/novaspivack/ugp-lean, 2026. https://doi.org/10.5281/zenodo.19554700. 117-module Lean 4 library (module count updated as of 2026-05-09). `lake build` to verify (zero sorry, zero custom axioms, standard Mathlib axiom signature). Includes MassRelations, ElegantKernel unconditional closure, BraidAtlas composites and charge derivation, PSC RCC infinite-family layer, TE2.2 scan certification. New in GTE.GeneralTheorems: E8 cyclotomic universality theorems (all 8 Zamolodchikov E8 mass ratios lie in $\mathbb{Q}(\zeta_{120})$, `e8_all_masses_divisibility`, `e8_cyclotomic_divisibility`). Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

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